

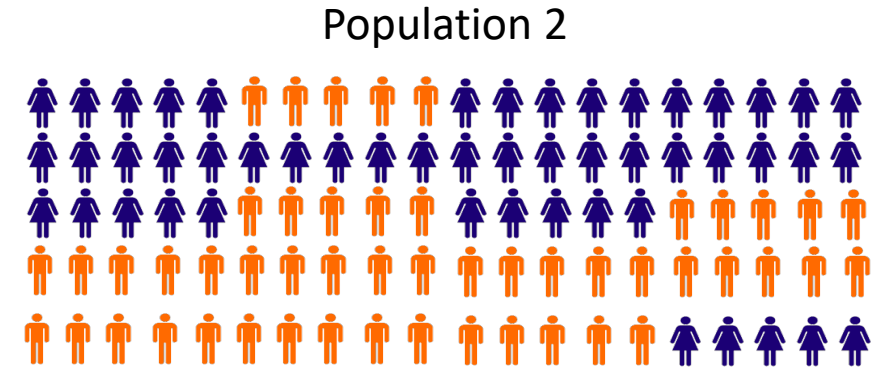
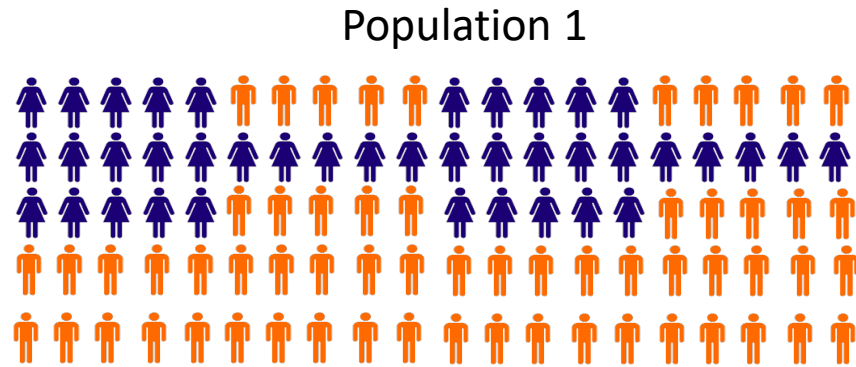


Survey Methods

3.1 - Power, precision and sample size

- The concept of power of a study
- Power, precision and sample size
- Principles of sample size calculation
- The 'paradox' of the preliminary data
- Myths about sample size calculations

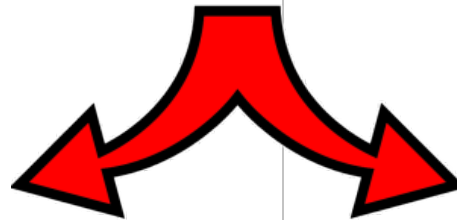
The concept of power: example



Research question: is the sex distribution the same in these two populations?

Hypothesis 1
(Null hypothesis)

Same proportion
males/females

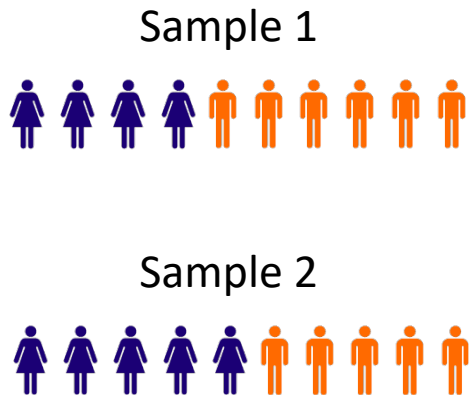
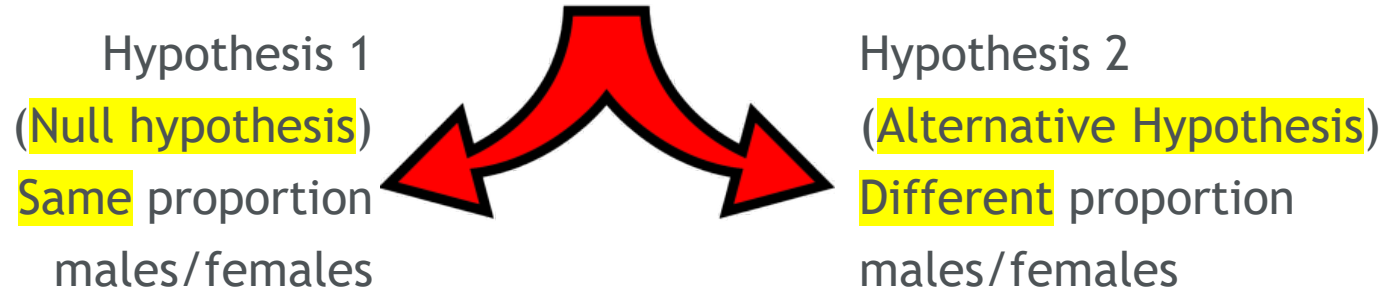


Hypothesis 2
(Alternative Hypothesis)

Different proportion
males/females

The concept of power: example

Research question: is the sex distribution the same in these two populations?



Statistics:
 $5/10 - 4/10 = 1/10 \neq 0$

Answer: reject null hypothesis

The concept of power

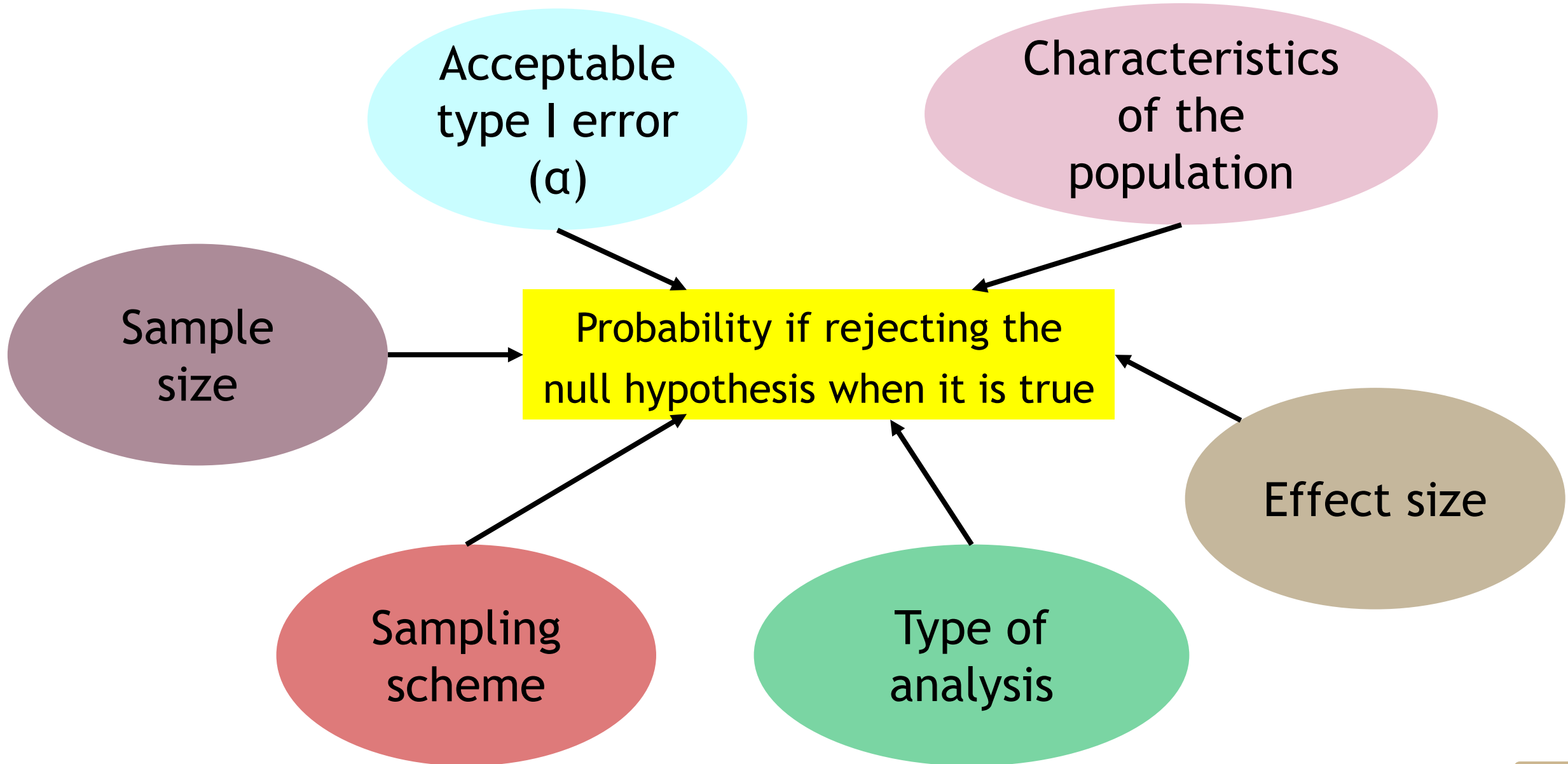
Hypothesis Testing Outcomes		REALITY	
		same proportion	Different proportion
RESEARCH	same proportion	 CORRECT DECISION	<i>Type II Error</i>  FALSE NEGATIVE
	Different proportion	<i>Type I Error</i>  FALSE POSITIVE	 CORRECT DECISION

The concept of power

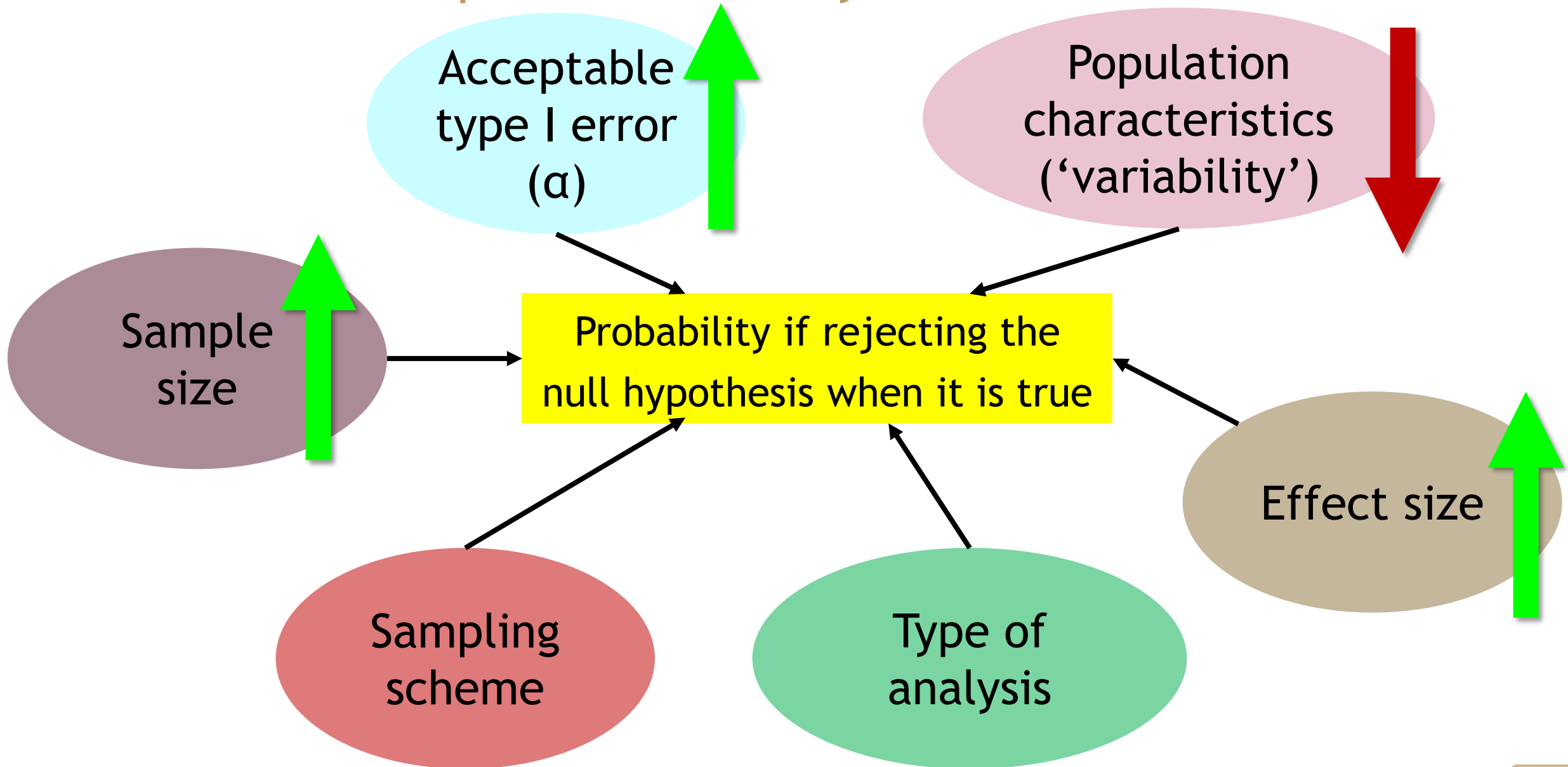
Hypothesis Testing Outcomes		REALITY	
		The Null Hypothesis is true	The Alternative Hypothesis is true
RESEARCH	The Null Hypothesis is true	$1-\alpha$  CORRECT DECISION	<i>Type II Error</i> β  FALSE NEGATIVE
	The Alternative Hypothesis is true	<i>Type I Error</i> α  FALSE POSITIVE	$1-\beta$  CORRECT DECISION

Power of a study

What affects the power of a study?



What affects the power of a study?



Example 1 : sample size for comparing two proportions

$$N = \frac{Z_{\alpha/2} \sqrt{2p(1-p)} + Z_{1-\beta} \sqrt{p_1(1-p_1)p_2}}{(p_1 - p_2)^2}$$

$Z_{\alpha/2}$ -1.96 for alpha 0.05

$Z_{1-\beta}$ - 0.84 for power 0.80

p1 and **p2** - proportion of event of interest (outcome) for group I and group II

p - (p1+p2) / 2

Effect size: finding a needle in a haystack



Effect size: finding a needle in a haystack



Example 2 : sample size for comparing two mean

$$N = \frac{(r + 1)(Z_{\alpha/2} + Z_{1-\beta})^2 \sigma^2}{rd^2}$$

r = n_1/n_2 - the ratio of sample size

σ - pooled standard deviation

d - difference of means of 2 groups

$Z_{1-\beta}$ - 0.84 for power 0.80

$Z_{\alpha/2}$ - 1.96 for alpha 0.05

Example 3 : sample size for single proportion (descriptive studies)

$$N = \frac{Z_{\alpha/2}^2 \times P \times (1 - p) \times D}{E^2}$$

power → precision



N - sample size

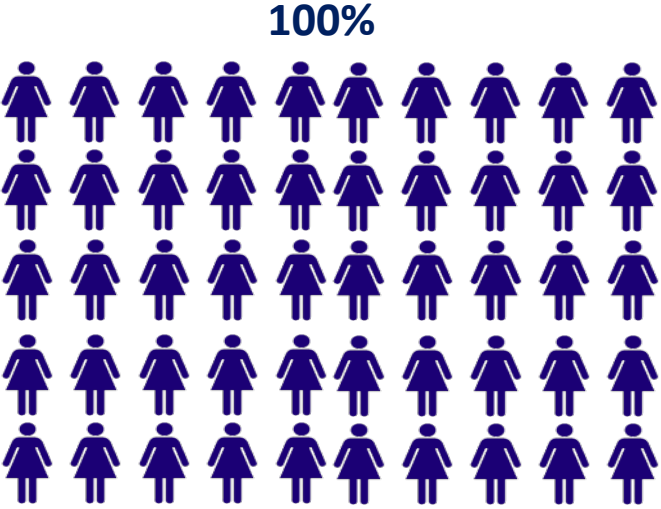
P - prevalence or proportion of event

E - precision (or margin of error) with which a researcher want to measure something

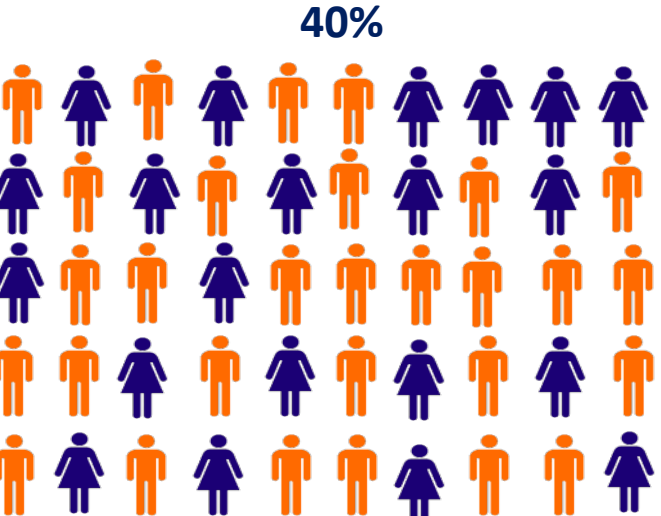
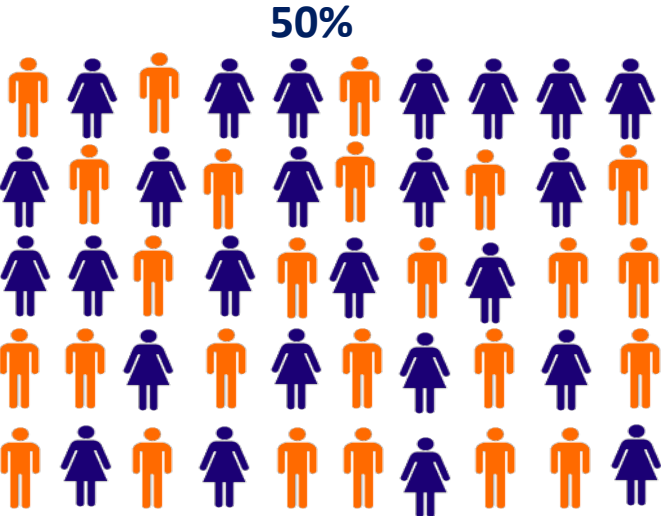
D - design effect reflects the sampling design used in the survey type of study. This is 1 for simple random sampling and higher values (usually 1 to 2) for other designs such as stratified, systematic, cluster random sampling

Z_{α/2} - 1.96 for alpha 0.05

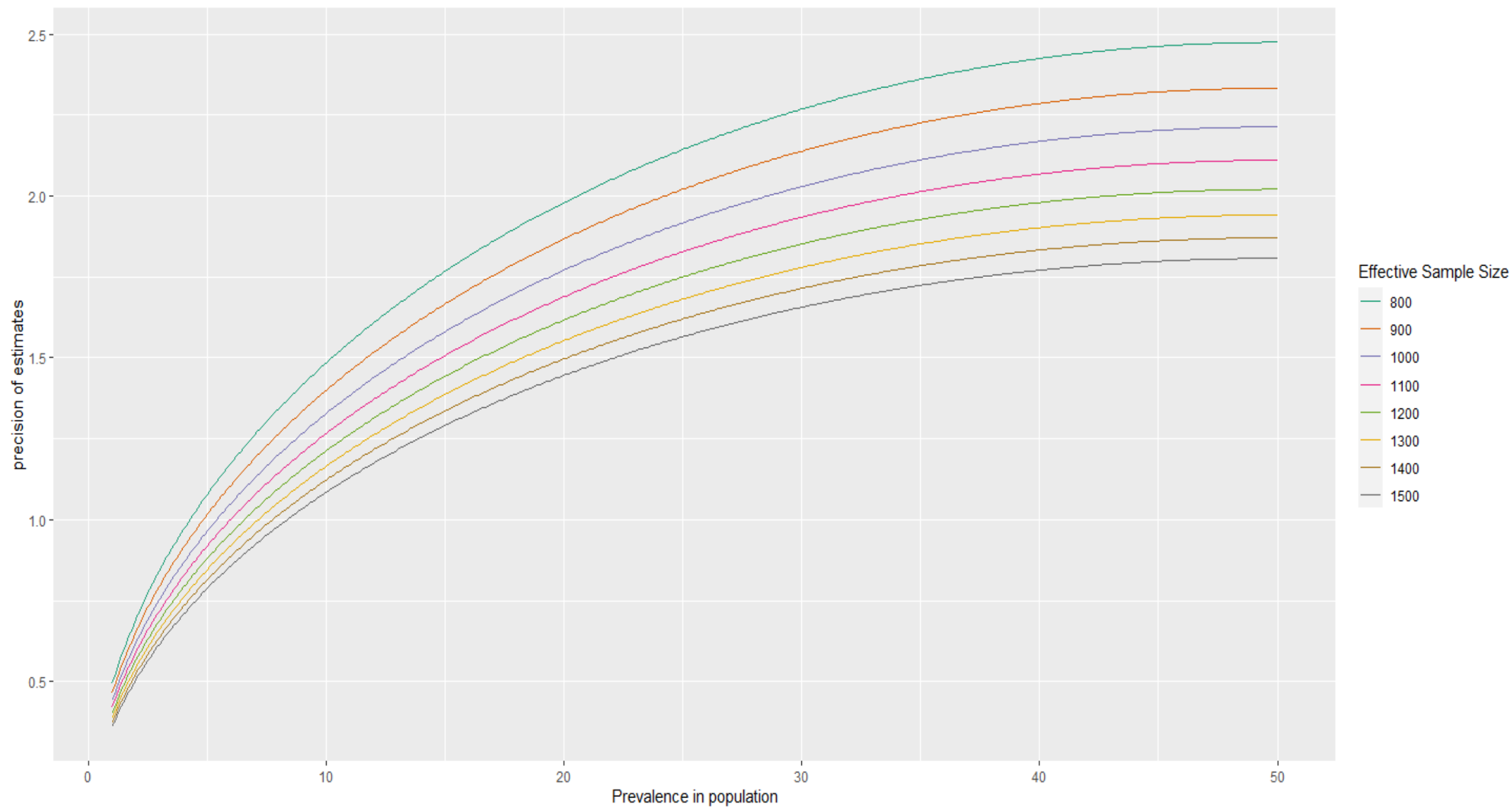
The effect of the baseline prevalence on the study power



$100\% - 90\% = 10\%$



$50\% - 40\% = 10\%$



Excel spreadsheet

SAMPLE SIZE BASIC CALCULATOR

PREVALENCE STUDY - BINOMIAL APPROXIMATION

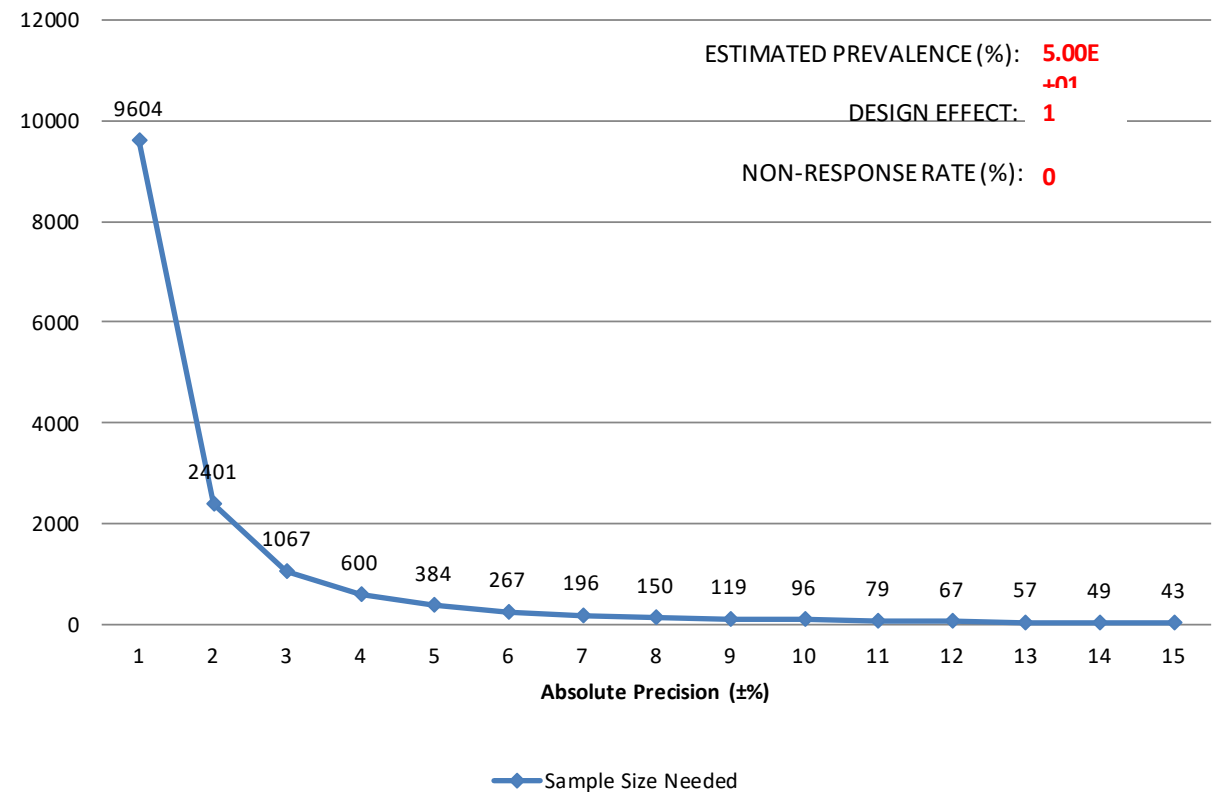
A) SAMPLE SIZE CALCULATION

REQUIRED ABSOLUTE PRECISION ($\pm$): d =	5.00E+00 %	
ESTIMATED PREVALENCE IN POPULATION (%): p =	5.00E+01 %	
ESTIMATED NON-RESPONSE RATE (%): nr =	0 %	
DESIGN EFFECT : df =	1	
EFFECTIVE SAMPLE SIZE : n1 =	384	(384.16)
SAMPLE SIZE NEEDED : n2 =	384	(384.16)

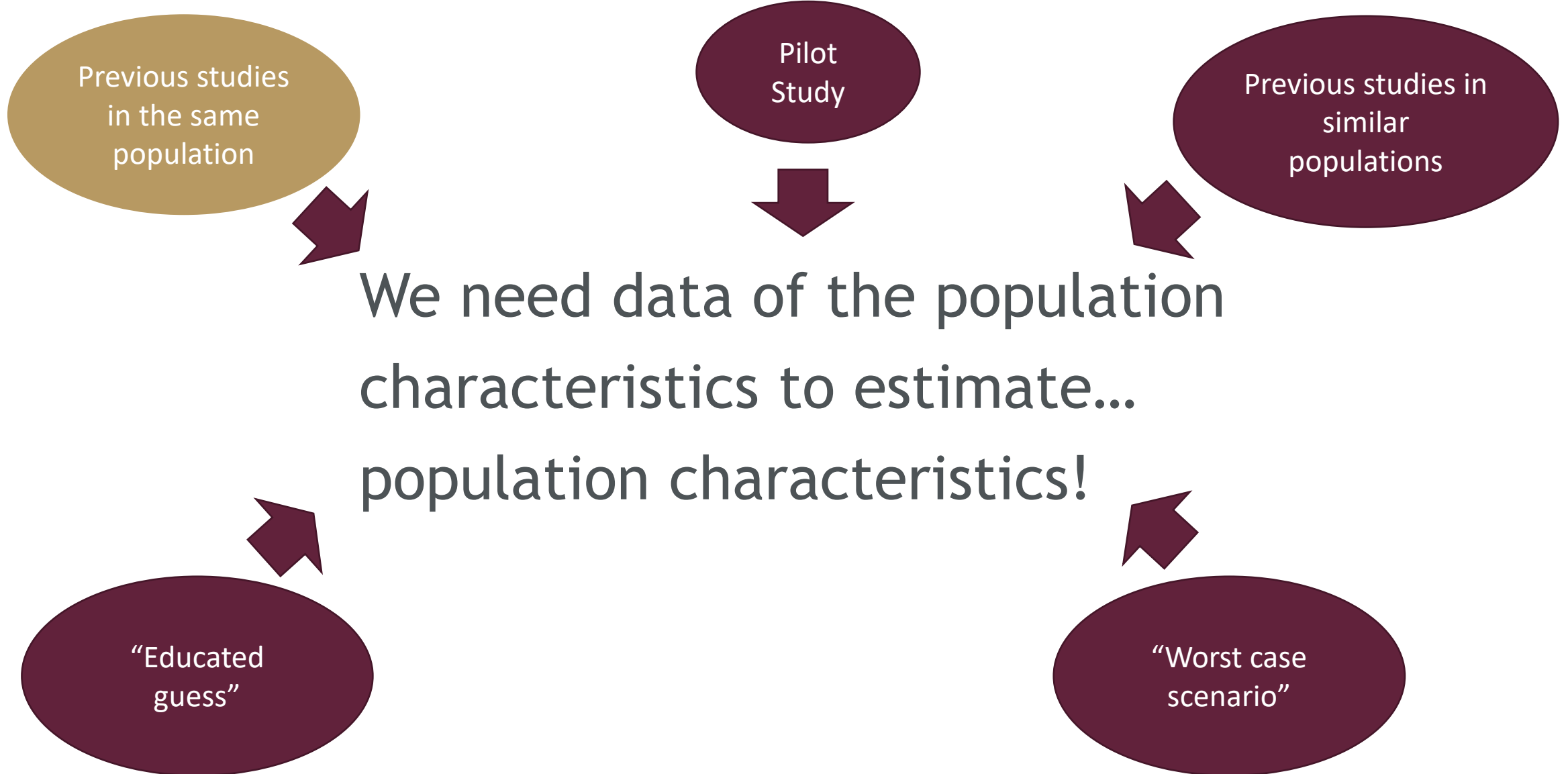
B) CALCULATION OF PRECISION

AVAILABLE SAMPLE SIZE : n2 =	2743	
ESTIMATED PREVALENCE IN POPULATION (%): p =	30 %	
ESTIMATED NON-RESPONSE RATE (%): nr =	20 %	
DESIGN EFFECT : df =	1.7	
EFFECTIVE SAMPLE SIZE : n1 =	1291	(1290.82)
ABSOLUTE PRECISION ($\pm$): d =	2.5	(2.50)

Sample Size Needed



Sourcing preliminary data



The effect of population size on the study power

Population Size	Margin of error (ME) (for CI 95%)	
	10%	5%
100	50	80
500	81	218
1000	88	278
10,000	96	370
100,000	96	383
1.000,000	97	384

(Adapted form Serdar et al. 2021:22)

Example

RESEARCH ARTICLE

Level of satisfaction and associated factors among patients attending outpatient departments of south Wollo health facilities, Ethiopia

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Objective 1

"This study sought to investigate the level of satisfaction expressed by participants regarding services provided by outpatient departments of selected health facilities in the south Wollo zone of Ethiopia and associated predictors"

Objective 2

Materials and methods

Study design and setting

A cross-sectional study was conducted among outpatient department customers at health facilities from May 13 to 25, 2019. The study was carried out in south Wollo, one of Amara National Regional State's fourteen administrative zones. Dessie city is 400 kilometers from Addis Ababa, the capital of the country and 480 kilometers from Bahirdar, the main city of Amara National Regional State. According to the information obtained from Amhara Regional State, Bureau of Finance and Economic Development (BOFED) the current estimated population is 3,095,457. The zone has a total of 11 primary hospitals, 131 health centers, 523 health posts, 134 private primary clinics (*BOFED report, 2020*).

Source and study population

All patients who visited the outpatient departments of the chosen health centers were considered the source population, and all patients who visited the outpatient departments of those ten health centers (from the five districts) during the study period were the study population.

Study variables

The study's dependent variable was level of patient-satisfaction. The independent variables were socio-demographic characteristics (age, religion, marital status, level of education, occupation), and outpatient service-related characteristics.

Inclusion and exclusion criteria

During the study period, patients over the age of 18 who visited health centers were included. Patients who were seriously ill at the time of data collection were excluded because they were not respond properly.



Sample size determination and sampling procedures

The following assumptions were used to calculate the required sample size using a single population proportion formula. Overall patient satisfaction in health care services is 80.1 percent [5], with an accuracy of 3% at a 95% confidence level and a 10% non-response rate.

$$n = \frac{Z_{1-\alpha/2}^2 p(1-p)}{d^2} = \frac{(1.96)^2 * 0.801(1-0.801)}{(0.05)^2} = 245$$

With a 10% non-response rate and a design effect of 2, the final sample size was $n = 540$.

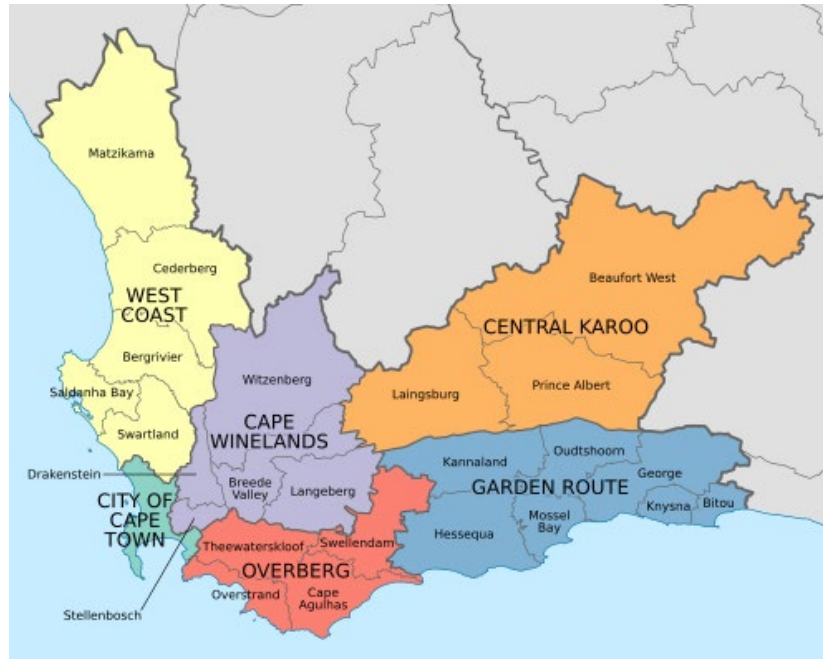
The sample size for the second objective was also computed using Epi-Info version 7 by applying double population proportion formula. The following assumptions: 95% confidence level, 80% power, 3.89 odds ratio, and 10% non-response were used to calculate the sample size for the second objective. However, the sample size was smaller than the sample size obtained for the first objective. Hence, the final sample size of the current study was 540 clients.

As a sampling technique, first multistage sampling technique was used. Five districts in the south Wollo zone were selected by lottery from a total of 24. (Wuchalie, Harbu, Bati, Mekane-selam and Albuko). Two health centers (HC) from each of these districts were selected again by lottery from the available HCs. Finally, the samples were drawn using a simple random sampling method, which involved taking the average daily patient flow from the adult outpatient department of the chosen HCs over the previous year's same month. Prior to data collection, the total samples were proportionally assigned to the HCs.

Objective 1:

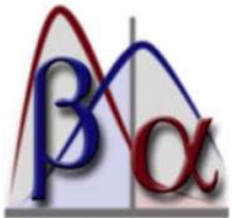
Objective 2

1. Which is the proportion of patients satisfied with the care received in PHC clinics in the Cape Winelands in 2021?
2. Is the proportion of patients satisfied with the care received in PHC clinics in the Cape Winelands in 2021 different from the proportion of satisfied patients in the West Coast in the same year?



Calculating sample size

Standard studies
Advanced Statistical
Training



STATA[®]



Standard studies
No Statistical Training
Complex studies
Basic/no Statistical Training

Standard studies
Basic Statistical
Training



ASK a statistician

<https://clincalc.com/stats/samplesize.aspx>

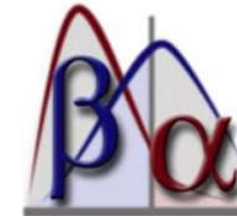
Preliminary estimates of
population characteristics

Margin of error/power

Acceptable risk

Resources

Preliminary sampling strategy



Summary questions

- Define power, precision, sample size
- Sample size calculations: choosing alpha, beta and precision
- What affects power/sample size?
- Source of preliminary data
- Minimum sample size and population size: what relationship?

References

Noordzij, Marlies, Giovanni Tripepi, Friedo W Dekker, Carmine Zoccali, Michael W Tanck, and Kitty J Jager. 2010. 'Sample Size Calculations: Basic Principles and Common Pitfalls'. *Nephrology Dialysis Transplantation* 25 (5): 1388–93. <https://doi.org/10.1093/ndt/gfp732>

Serdar, Ceyhan Ceran, Murat Cihan, Doğan Yücel, and Muhittin A. Serdar. 2021. 'Sample Size, Power and Effect Size Revisited: Simplified and Practical Approaches in Pre-Clinical, Clinical and Laboratory Studies'. *Biochemia Medica* 31 (1): 010502. <https://doi.org/10.11613/BM.2021.010502>.

Other resources

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Ayele WM, Ewunetu A, Chanie MG (2022) Level of satisfaction and associated factors among patients attending outpatient departments of south Wollo health facilities, Ethiopia. *PLOS Global Public Health* 2(7): e0000761.

<https://doi.org/10.1371/journal.pgph.0000761>



Thank you

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